Local quantum-field probes: causal measurements and reliable resource harvesting

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Abstract

Localized quantum probes offer an operational route from quantum-field correlations to finite systems whose resources can be measured. This critical narrative review examines the conditions under which that route supports a reliable harvesting claim. We organize selected work around three comparisons: outcome probabilities versus measurement instruments, correlation transfer versus signal-mediated generation, and resource witnesses versus attainable task performance. Detector, algebraic system–probe, and event-probability approaches are compared by their outputs and causal assumptions. Data processing constrains accessible correlations, while the full spacetime support of an interaction determines the field region to which such bounds apply. Worked examples illustrate a strictly spacelike pointlike limit, the limits of residual-negativity indicators for mixed-state multipartite entanglement, and the difference between structural channel invariance and a background-specific effect. Scalar, Dirac, and electromagnetic probes are compared through their coupled observables rather than temperature alone. We close by assessing the evidence supplied by optical correlation measurements and proposed detector platforms, and by specifying the measurements and approximation checks still needed for resource certification.

Keywords: relativistic quantum information; quantum field theory; local measurements; entanglement harvesting; quantum resources; causality; resource certification

1 Introduction

Relativistic quantum information asks which information-processing operations localized laboratories can implement when their probes interact with quantum fields. A central example is entanglement harvesting: initially uncorrelated probes may become entangled after local interactions with a correlated field. The calculation connects a field-theoretic object to a finite detector state, but that connection involves several choices. A field correlation, a probe response, an entanglement witness, and a useful protocol output are different objects. Their relation depends on the interaction, its complete spacetime support, the readout, and the approximation used [1–4].

This review addresses one question: when does localized access to a quantum field justify a claim of resource extraction? The question connects local measurement theory with harvesting and certification. It also explains why superficially similar resource curves need not be comparable. A finite probe has a reduced density matrix; a continuum field region is described by a local observable algebra. A nonzero correlation at spacelike separation is compatible with vanishing causal influence. Equal thermal response temperatures need not imply equal nonlocal correlators. These distinctions are operational requirements for comparing models, rather than obstacles to using detector models.

Existing reviews establish the broader setting. Peres and Terno survey quantum information and relativity; Hu, Lin, and Louko discuss detector dynamics and information protocols; Bukhari and Wang review atom–field dynamics in curved spacetime. Papageorgiou and Fraser

examine the local-measurement problem in QFT, while general quantum-resource reviews provide conversion rules and certification tools [1, 2, 5–8]. Table 1 identifies the comparison developed here. It is a statement of scope, not a claim that earlier reviews omitted every issue in its final column.

Table 1: Positioning of the present synthesis relative to complementary reviews.

Literature strand	Established emphasis	Comparison developed here
Quantum information and relativity [1]	Relativistic subsystem descriptions and information tasks	Which field and probe subsystems a harvesting claim actually concerns.
Detector dynamics and curved-spacetime atom–field physics [2, 5]	Response, open dynamics, and background effects	Which differences survive matching of coupled observables, supports, and approximations.
Local QFT measurement theory [6]	Causal admissibility and measurement frameworks	Which framework supplies probabilities, conditional states, or a final probe resource.
Quantum-resource theory [7, 8]	Witnesses, monotones, and conversions	Which detector-state indicators support certification under the stated measurement scenario.

The contribution is a critical synthesis of these comparisons, supported by explicit diagnostics. We distinguish constraints that follow from general channel arguments from conclusions that require a detector calculation. We track the entire interaction support when assigning regional correlation bounds. We also distinguish mixed-state entanglement certification from residual combinations of bipartite measures, and structural invariance under local noise from a specifically relativistic protection effect. These examples do not define a new resource theory or establish a universal extraction efficiency.

This narrative review considers selected foundational and recent studies through 2 November 2026. It does not claim an exhaustive search or a systematic-review protocol. Inclusion is guided by relevance to local measurements, harvesting, certification, and closely connected tasks. Preprints are identified where used. Detailed calculations are separated from the literature synthesis; their assumptions and verification limits are stated in the supplement. Satellite-network engineering, independent quantum-clock proposals, and gravity-mediated transfer are outside the present scope.

Section 2 fixes the comparison criteria. Sections 3–5 treat causal measurements, accessible correlations, and harvesting. Section 6 examines resource certification, including mixed-state multipartite counterexamples. Sections 7 and 8 give two tests of utility. Section 9 compares backgrounds and coupled observables; Sec. 10 assesses experimental evidence and priorities.

2 A common operational framework

We use units $c = \hbar = k_B = 1$ and metric signature $(-,+,+,+)$ unless stated otherwise. For a real scalar field in a state ρ_ϕ, the Wightman two-point function is

$$W(x,x') = \mathrm{Tr}\left[\rho_\phi\, \hat{\phi}(x)\hat{\phi}(x')\right]. \tag{1}$$

It characterizes field correlations and generally need not vanish when x and x' are spacelike separated. By contrast, microcausality requires

$$[\hat{\phi}(x), \hat{\phi}(x')] = 0 \qquad \text{for spacelike-separated } x, x'. \tag{2}$$

For a free scalar field, the commutator is a c-number distribution and is closely related to the causal propagator. The two functions answer different questions: $W(x,x')$ describes correlations,

whereas the commutator determines whether local field operations can influence one another through causal propagation.

This distinction is central to RQI. Nonzero spacelike correlations do not by themselves permit superluminal signalling. Conversely, a vanishing commutator does not imply that the joint state of two localized regions is uncorrelated. Detector-based harvesting exploits precisely this separation: localized probes can sometimes acquire entanglement from field correlations even when their interaction regions are strictly spacelike separated, provided the protocol does not allow communication between the probes [9].

A resource theory is specified by a set of free states, a class of free operations, and a way to quantify or order states according to the resource of interest. This structure matters because a numerical witness is not automatically a resource monotone, and two different resource measures need not induce the same ordering of states. The distinction is familiar from the literature on coherence and geometric quantum discord, where the measure, basis, and allowed operations affect the interpretation of the resource [8].

For relativistic QFT, an additional layer is required: the physically admissible operations must be compatible with spacetime localization and causal structure. The relevant operational chain can be summarized as

$$\begin{aligned} \text{field state} &\longrightarrow \text{localized interaction} \\ &\longrightarrow \text{measurement record or probe state} \\ &\longrightarrow \text{resource witness} \longrightarrow \text{information task.} \end{aligned} \tag{3}$$

Each arrow requires justification. In particular, a nonzero field correlation does not guarantee that a chosen detector can access it; a detector response does not automatically certify a resource transfer; and a resource witness does not by itself establish a task advantage.

The five stages can be used as a checklist when reading or comparing a result. For example, a nonzero Wightman function establishes field correlation, not a communication channel. A positive detector negativity establishes entanglement in the chosen reduced probe state, but does not by itself identify whether the entanglement was harvested, transferred, or generated through a causal interaction. A metrological gain requires a specified parameter, measurement strategy, resource budget, and comparison benchmark. The minimum evidence required at each stage is summarized in Table 2.

Table 2: Five operational stages for assessing relativistic quantum-information claims.

Stage	Central question	Minimum information needed for a meaningful claim
Resource existence	Which state or field region carries the resource?	State, subsystem or local algebra, resource definition, and assumptions on the field state.
Operational accessibility	Can a localized apparatus access it?	Interaction Hamiltonian or local operation, spacetime support, detector profile, and causal relations.
Measurement reliability	Can the resource be certified?	POVM or measurement record, noise model, finite resolution, estimator, and statistical uncertainty.
Generation or transfer	What physical process produces the final resource?	Initial resources, interaction region, perturbative order or exact dynamics, and a causal attribution test.
Task utility	Does the resource improve a task?	Encoding and decoding protocol, task figure of merit, energy/time budget, and a matched baseline.

The checklist fixes the comparison criteria used below; it does not define a new resource

theory.

3 Local measurements and causal implementability

3.1 Probabilities, instruments, and local apparatus

Applying an ideal projection postulate to an arbitrary QFT observable can produce Sorkin-type impossible-measurement scenarios. The issue is which operations a spacetime-localized apparatus can implement, not simply whether two observables commute [6, 10, 11]. A POVM specifies outcome probabilities; an instrument additionally specifies conditional state changes [12, 13]. Two constructions can agree on a probability without inducing the same back-action.

A two-level Unruh–DeWitt (UDW) detector provides a tractable apparatus model. For a worldline $x(\tau)$,

$$H_I(\tau) = \lambda\chi(\tau)\mu(\tau)\hat{\phi}[x(\tau)]. \tag{4}$$

Finite-size detectors replace the point evaluation by a spatial smearing. The preparation, gap, switching, profile, and readout determine the induced field measurement and the final probe state. Detector-based measurement theory makes that induced POVM and update explicit [14]. Fully relativistic localized-field probes reproduce leading detector predictions in specified regimes, supporting the approximation without identifying all probe models or perturbative orders [9, 15, 16].

The Fewster–Verch framework instead couples target and probe QFTs in a compact region and induces observables and instruments through a scattering morphism [17]. Local couplings preserve causal consistency in the system–probe setting [18]. Constructive results for linear scalar fields include asymptotic local-probe schemes and specified smeared-field measurements [19, 20]. Their assumptions do not amount to a classification of every instrument in interacting QFT.

3.2 Event probabilities and causal composition

Quantum temporal probabilities (QTP) retain detection time as an outcome. Their leading event density has the schematic form

$$P(x,\lambda) = \int d^4y\, R^{ab}(y,\lambda) G_{ab}(x-y/2, x+y/2), \tag{5}$$

where R^{ab} describes apparatus response and coarse-graining. Joint-event probabilities involve selected unequal-time higher-point correlators. UDW response and Glauber photodetection arise in limiting regimes, rather than defining interchangeable measurements [21–24]. Event-hierarchy proposals examine quantum features of multi-time probabilities; these indicators concern a specified hierarchy, not automatically the entanglement of a reduced detector state [25].

Other approaches emphasize the composition of localized operations. Positive-formalism constructions address positivity and composition for specified observable classes; a multi-observable expectation is not itself a joint outcome distribution [26, 27]. Local S-matrix constructions test causal factorization and stronger conditions on operations, including pointer back-action in a specified model [28]. These results complement explicit detector instruments. Their different output objects are summarized in Table 3.

A useful benchmark fixes one field state, compact interaction supports, detector preparation, smearing, switching, and pointer readout. It first compares integrated single-probe probabilities, then joint event statistics and selective updates. Only an explicit probe instrument supplies the reduced state needed for detector entanglement. Supplementary Sec. S1 gives this benchmark, including the distinction between coincidence covariance and negativity. It is a proposed comparison protocol, not an already completed equivalence proof between the frameworks.

Table 3: Operational outputs of complementary measurement constructions.

Construction	Direct output	Additional requirement for a resource claim
Explicit detector instrument [14]	Probe probabilities, induced field update, final probe state	Controlled dynamics, readout model, and a witness on that state.
FV system–probe scheme [17]	Induced local observables and instruments	A specified probe preparation and implementation of the relevant readout.
QTP [23]	Time-resolved and joint event statistics	An explicit link between those statistics and the proposed resource criterion.
Positive formalism / local S matrix [27, 28]	Probabilistic composition / admissible local operations	A matching apparatus and outcome functional; causal admissibility alone does not specify detector negativity.

3.3 Reliability of certification

A witness computed from an ideal reduced state certifies that model. Experimental certification also requires readout calibration and statistical uncertainty. If $p(r|\rho) = \mathrm{Tr}(E_r\rho)$, the map to observed counts includes detector preparation, coupling, switching, readout noise, and postselection. A lower confidence bound on a witness should remain positive over the calibrated uncertainty set. A resource monotone additionally obeys non-increase under a specified free-operation class; being a useful numerical indicator is not sufficient.

Locality and detector reliability impose independent restrictions. Bounds relating sensitivity to vacuum dark counts depend on the apparatus and class of excitations considered [29]; they do not replace a resource-specific error analysis. Abrupt switching and finite smearing change ultraviolet behavior and the balance of local noise against nonlocal coherence [4, 30–32]. A small positive leading-order witness is especially vulnerable near its zero boundary. Complete causal-support tests, numerical convergence, and control of omitted perturbative orders are therefore reported separately below.

4 Operational accessibility of field resources

A local measurement framework specifies which operations are physically available. The next question is how much of the field-level information can be accessed by those operations. This requires a comparison between local field algebras, regional correlations, detector channels, and the causal support of the interaction. The following results provide complementary bounds rather than a single universal measure of accessible resource.

The section combines established algebraic results with selected quantitative analyses. The channel-contraction and finite-mode Jaynes–Cummings calculations are illustrative benchmarks developed here to test what data processing can and cannot establish. The pointlike Unruh–DeWitt scan is a separate model-specific detector calculation. None of these examples proves a universal bound for the full continuum detector channel.

4.1 Regional correlations and their domain of applicability

Field mutual information can be defined using algebraic relative entropy rather than a finite-dimensional tensor factorization. A normal product reference state and appropriate finiteness assumptions are required. For gauge fields the local algebra, shared centre, and superselection sectors must also be specified. These choices precede any comparison with detector correlations.

Modular-flow calculations and exact free-field relations provide useful regional information for specified geometries [33–35]. An approximation for CFT balls, however, is not an exact

mutual information for an arbitrary detector world tube. The detailed ball formulas and a clearly identified approximate application are placed in Supplementary Sec. S7. The main argument below requires only data processing and the correct localization of the detector channels.

4.2 A data-processing bound on detector-accessible entanglement

The field-to-detector distinction can be quantified without assuming a spatial tensor-product decomposition of the field. Let $\mathcal{A}_A$ and $\mathcal{A}_B$ be commuting local algebras for spacelike separated regions, and assume that a normal product reference state exists (for example, under a suitable split or finite-resolution prescription). Define

$$I_{\mathrm{F}}(A:B) = S(\omega|_{\mathcal{A}_A\vee\mathcal{A}_B} \,\|\, \omega|_{\mathcal{A}_A} \otimes \omega|_{\mathcal{A}_B}) \,. \tag{6}$$

In gauge theories, the local algebras, shared centre, and superselection sectors must be specified first. The Araki relative entropy may still be infinite; modular p-nuclearity for some $0 < p < 1$ is a sufficient finiteness condition in the class studied by Panebianco and Wegener (arXiv:2110.05823). Thus the following bound is quantitatively informative only when $I_{\mathrm{F}} < \infty$, and regulated results must state their regulator and limiting procedure.

Suppose D_A and D_B interact locally with $\mathcal{A}_A$ and $\mathcal{A}_B$, respectively, start in a product state, and share no nonlocal ancilla or direct interaction. The induced local channels and monotonicity of relative entropy give

$$I(D_A:D_B) \le I_{\mathrm{F}}(A:B). \tag{7}$$

For the detector state, define the relative entropy of entanglement

$$E_R(\rho_{AB}) = \inf_{\sigma\in\mathrm{SEP}} S(\rho_{AB}\|\sigma). \tag{8}$$

Since $\rho_A\otimes\rho_B$ is separable, $E_R(\rho_{AB}) \le I(D_A:D_B)$. Pinsker's inequality and $\|X^{T_B}\|_1 \le d_B\|X\|_1$ then imply

$$\mathcal{N}(\rho_{AB}) \le d_B\sqrt{\frac{E_R(\rho_{AB})}{2}}, \tag{9}$$

where $\mathcal{N} = (\|\rho_{AB}^{T_B}\|_1 - 1)/2$. All entropies in these bounds use natural logarithms; base-two entropies require the corresponding factor of $\ln 2$. Combining the bounds yields

$$\mathcal{N}(\rho_{AB}) \le \min\left\{\frac{\min(d_A,d_B)-1}{2}, d_B\sqrt{\frac{I_{\mathrm{F}}(A:B)}{2}}\right\}, \tag{10}$$

or, for two qubits,

$$\mathcal{N}(\rho_{AB}) \le \min\left\{\frac{1}{2}, \sqrt{2I_{\mathrm{F}}(A:B)}\right\}. \tag{11}$$

The inequality is rigorous but generally not tight; it follows from data processing and finite-dimensional norm bounds, not from an optimal harvesting construction.

To obtain the intermediate estimate, choose a separable σ approaching the infimum in E_R. Pinsker's inequality gives $\|\rho-\sigma\|_1 \le \sqrt{2S(\rho\|\sigma)}$. Since $\|\sigma^{T_B}\|_1 = 1$ and partial transposition obeys $\|X^{T_B}\|_1 \le d_B\|X\|_1$, the triangle inequality yields $\mathcal{N}(\rho) \le (d_B/2)\|\rho-\sigma\|_1$. Taking the infimum gives Eq. (9). The dimension-dependent cap in Eq. (10) is the maximum possible negativity for the detector dimensions.

No positive lower bound on detector entanglement follows from I_{F} alone: a probe may miss the correlated modes, have zero coupling, or induce an entanglement-breaking channel. In a specified perturbative Unruh–DeWitt model, the condition $|\mathcal{M}| - \sqrt{P_A P_B} > 0$ is detector-specific, not a field-only lower bound. Local noise can only reduce the unconditioned detector correlations; a postselected branch must be weighted by its success probability when assessing

average yield. A recent leaky-cavity preprint reporting no-click conditional concurrence is a different, single-mode setting (arXiv:2608.13449).

For gauge fields, applying the argument requires a gauge-invariant local algebra and a treatment of boundary centres or superselection sectors. A two-dimensional lattice-gauge construction supplies a sector-wise tensor-product structure and a generalized LOCC argument, but its discrete decomposition does not automatically extend to type-III continuum algebras (arXiv:2512.19806). The Maxwell extension is therefore conditional on the chosen algebra and finite-resolution prescription.

Three levels must be distinguished. Field mutual information $I_{\mathrm{F}}(A : B)$ measures total correlations between local field algebras, including classical and quantum parts. Detector mutual information $I(D_A : D_B)$ measures total correlations in the final probe state and depends on the chosen preparation and local channels. Detector entanglement, quantified here by E_R or negativity, is more restrictive and is not the entanglement of the full field regions. The data-processing and separable-state bounds relate these quantities but do not make them equal: $I_{\mathrm{F}} > 0$ does not guarantee nonzero detector negativity.

A complementary Gaussian-mode result bounds the entanglement extractable by a fixed number of localized bosonic probes using the most mixed normal modes of the reduced covariance matrix [36]. It concerns mode selection in a pure Gaussian state, not algebraic mutual information or a general detector-channel bound.

4.3 Channel dependence and finite-resource bounds

Data processing is universal, whereas a strict contraction factor is channel dependent. For local probe channels $\mathcal{E}_A, \mathcal{E}_B$, a bound of the form

$$I(D_A : D_B) \leq \eta_A \eta_B I_{\mathrm{F}}(A : B), \qquad 0 \leq \eta_A, \eta_B \leq 1, \tag{12}$$

is useful only if the coefficients are established for the same input class and regulator as the field mutual information. Complete contraction and quantum Doeblin methods give sufficient conditions in specified noisy-channel settings [37, 38]. They do not assign a detector-independent extraction efficiency to a relativistic field.

The ideal Jaynes–Cummings channel illustrates the obstruction: rare, high-energy Fock inputs can preserve an encoded bit almost perfectly even under a fixed mean-energy budget. An average-energy constraint need not imply a strictly smaller worst-case mutual-information contraction, a mechanism related to classical average-cost counterexamples [39]. A hard cutoff, explicitly calibrated noise, or a state-dependent one-sided data-processing estimate can be more informative. Supplementary Sec. S11 gives the detailed argument; the finite-mode transfer example in Supplementary Sec. S5 illustrates the difference between field mutual information, detector mutual information, and detector negativity [40]. These examples explain the limits of an upper bound; they are not continuum harvesting protocols.

A two-mode squeezed vacuum coupled locally to two ground-state qubits through resonant Jaynes–Cummings channels provides a finite-dimensional check of the field-to-detector bounds. The exact detector state has an X form, and its negativity can be evaluated from convergent Fock sums. For example, at $\theta = \pi/2$ and $r = 0.4$, the field mutual information is $I_{\mathrm{F}} = 0.96490$ nats, the detector mutual information is 0.69518 nats, and the detector negativity is 0.32103. In the weak-squeezing limit, $\mathcal{N} = r \sin^2\theta + O(r^2)$, whereas $I_{\mathrm{F}} = 2r^2[1 - \ln(r^2)] + O(r^4|\ln r|)$; this explains why a general mutual-information bound can be loose for weak inputs. A one-sided, state-dependent data-processing estimate reduces the field-derived bound by about 29% at $r = 0.05$ and 0.10 for $\theta = \pi/4$. The finite-mode benchmark is not a spacelike-separated relativistic harvesting calculation. Its exact X-state expressions, local-population bound, Fock-cutoff convergence, figure, and data table are retained in Supplementary Sec. S5 [40].

4.4 The pointlike limit of strictly spacelike detectors

A compact-support calculation for two identical static UDW detectors in the four-dimensional massless scalar vacuum shows that harvested entanglement can survive the pointlike limit when the switching duration and detector separation are fixed. The complete interaction supports obey $L - 2\sigma > 2T$ throughout the scan. At $T = 1$, $\Omega T = 9$, and $L/T = 2.033333\ldots$, the pointlike response coefficients are $p_0 = 2.13325790 \times 10^{-6}$ and $|m_0| = 3.57653182 \times 10^{-6}$, giving $\lim_{\sigma\to 0} \mathcal{N}_\sigma/\lambda^2 = 1.44327392 \times 10^{-6} + O(\lambda^2)$. Thus the positive leading-order entangling gap is not caused by a finite spatial detector radius. This conclusion concerns the detector state for the specified field, switching, and weak-coupling regime; it does not determine the mutual information of shrinking equal-time field regions. The full compact profiles, small-size expansion, quadrature checks, and figure are retained in Supplementary Sec. S6.

4.5 Which field region controls the data-processing bound?

The pointlike detector limit raises a geometric issue in applying the field-to-detector data-processing bound. The detector interaction is not supported on an equal-time ball of radius σ. It is supported on the spacetime world tube

$$\mathcal{W}_A = \{(t, \mathbf{x}) : |t| < T,\ |\mathbf{x} - \mathbf{x}_A| < \sigma\}, \tag{13}$$

and similarly for detector B. A centred causal double cone containing this nominal open world tube is the domain of dependence of the equal-time ball of radius

$$R = T + \sigma. \tag{14}$$

Indeed, the double cone $D(B_R)$ is characterized by $|t| + |\mathbf{x} - \mathbf{x}_A| < R$, which contains every point of $\mathcal{W}_A$ when $R \geq T + \sigma$. The smooth interaction support is taken compactly inside the nominal world tube; a closed-support convention can instead use an arbitrarily small enlargement of R. The strict-spacelike condition $L - 2\sigma > 2T$ is exactly $L > 2R$. Thus the two enclosing double cones remain spacelike separated throughout the pointlike limit, with $R \to T$ rather than $R \to 0$.

The local detector channels are therefore localized in the algebras of these enclosing double cones, not in the algebras of the much smaller equal-time balls of radius σ. Under the split/finite-resolution assumptions used above, the appropriate data-processing statement is

$$I(D_A : D_B) \leq I_{\mathrm{F}}(D(B_{T+\sigma})_A : D(B_{T+\sigma})_B)\,. \tag{15}$$

By the time-slice property, the algebra of a causal double cone can be represented on its equal-time Cauchy ball. This permits the use of spatial-ball mutual-information methods, but with radius $T + \sigma$, not σ.

This distinction matters because the known large-separation expansion for two equal-time balls in the four-dimensional massless scalar vacuum has the leading scaling

$$I_{\mathrm{F}}(B_R^A : B_R^B) \sim C_{\mathrm{sc}} \left(\frac{R}{L}\right)^4, \qquad R/L \ll 1, \tag{16}$$

where C_{sc} depends on the field theory and the region convention [33, 41]. This is the small-region expansion of the field mutual information; it is not the scaling of the detector negativity. Substituting the correct enclosing radius gives

$$I_{\mathrm{F}} \sim C_{\mathrm{sc}} \left(\frac{T + \sigma}{L}\right)^4 \longrightarrow C_{\mathrm{sc}} \left(\frac{T}{L}\right)^4 \quad (\sigma \to 0), \tag{17}$$

provided the large-separation regime applies. The field mutual-information budget therefore does not vanish merely because the detector's spatial radius tends to zero while its interaction duration remains fixed.

For the numerical harvesting point used above, $L/T = 2.033333\ldots$ and the pointlike enclosing radius is $R/T = 1$. The equal-sphere cross ratio is $z = 4R^2/L^2 \simeq 0.9675$, so the large-separation expansion in Eq. (16) is not quantitatively controlled there. It would be incorrect to insert that asymptotic formula as a numerical value for the bound. A direct Gaussian-field or regulated algebraic calculation of the mutual information of the two nearly touching double cones is required.

The geometric conclusion is independent of the remaining numerical uncertainty: the apparent conflict between a vanishing mutual information of shrinking equal-time balls and a finite detector pointlike negativity results from assigning the detector channel to a region that does not contain its full spacetime support. The correct enclosing region retains a finite temporal scale. An approximate modular-flow estimate is retained in Supplementary Sec. S7; a regulator-controlled calculation of the exact four-dimensional scalar mutual information remains necessary.

The modular-flow estimates in Supplementary Sec. S7 illustrate the looseness of this bound. At the near-contact benchmark, the available ansatz estimates exceed the two-qubit threshold $I_{\mathrm{F}} = 1/8$, so inserting them yields only the dimensional cap. At larger separation they suggest a more informative comparison, but an ansatz is neither an upper bound on the exact field mutual information nor a certified error interval. The geometric localization argument above is independent of that approximation; quantitative field-derived negativity bounds require a controlled value or rigorous upper bound for I_{F}.

5 Entanglement harvesting and causal attribution

5.1 The detector-state criterion

Harvesting couples initially uncorrelated probes locally to a field and tests their final reduced state. Vacuum-extraction proposals and detector calculations established that pre-existing field correlations can be accessed without signal exchange between the complete interaction regions [3, 4, 42]. For initially ground-state two-level probes, the potentially negative eigenvalue of the relevant partial-transpose block is

$$\lambda_-^{(2)} = \frac{P_A + P_B - \sqrt{(P_A - P_B)^2 + 4|\mathcal{M}|^2}}{2}. \tag{18}$$

With $\mathcal{N} = (\|\rho^{T_B}\|_1 - 1)/2$, the leading negativity is therefore

$$\mathcal{N}^{(2)} = \max\left\{0, \frac{\sqrt{(P_A - P_B)^2 + 4|\mathcal{M}|^2} - (P_A + P_B)}{2}\right\}. \tag{19}$$

The condition $|\mathcal{M}| > \sqrt{P_A P_B}$ is a threshold, not the general value of negativity. For identical probes it reduces to $\max(0, |\mathcal{M}| - P)$. The local probabilities and nonlocal coherence must be evaluated at the same order; omitted terms can decide the sign near threshold.

The quantities are interaction dependent. Switching, smearing, derivative coupling, detector dimension, and initial preparation change the sampled kernels. Antiparallel and parallel acceleration can consequently produce different trends in specified setups [43]. Localized relativistic QFT probes reproduce leading harvesting predictions under stated assumptions [9]; Gaussian-mode extraction bounds instead concern mode selection in a specified covariance-matrix setting [36]. Neither establishes a universal detector-independent harvesting yield.

5.2 Full support and the origin of the nonlocal amplitude

For strictly spacelike interaction supports, microcausality excludes inter-probe signalling. If the supports are causally connected, the final entanglement can contain a signal-mediated contribution. For free scalar models at leading order, a commutator–anticommutator decomposition separates state-independent causal response from state-dependent correlations [44]. The amplitudes can interfere:

$$|\mathcal{M}|^2 = |\mathcal{M}_{\mathrm{corr}}|^2 + |\mathcal{M}_{\mathrm{comm}}|^2 + 2\,\mathrm{Re}(\mathcal{M}_{\mathrm{corr}}\mathcal{M}^*_{\mathrm{comm}}). \tag{20}$$

Negativity is nonlinear in the total amplitude and local noise. It cannot in general be partitioned into two positive independent yields. Constructive and destructive interference are model-dependent possibilities [45, 46]. Supplementary Sec. S2 retains a conformally coupled de Sitter example, with its noncompact-profile and perturbative qualifications.

Gaussian profiles have noncompact tails. Centre separation or a small communication-mediated-entanglement estimator does not prove exact spacelike separation of the interactions. Compact support establishes the strict condition; a tail approximation instead requires an explicit error or signalling budget. The compact benchmark in Sec. 4.4 supplies an example of the former. Covariance of a specified perturbative calculation and absence of signal exchange are separate checks [47].

Table 4: Causal classifications and the evidence they require.

Regime	Required check	Interpretation
Strict spacelike extraction	Every pair of points in the two compact interaction supports is spacelike	Inter-probe signal exchange is excluded; the probe-state witness still needs approximation control.
Effective separation with tails	Tail contribution or signalling error is bounded for the stated profiles	Extraction is assessed within that error budget, without asserting exact spacelike support.
Causal contact	Full local and nonlocal amplitudes, including interference, are retained	A positive final witness alone does not distinguish harvesting from signal-mediated generation.
Postselected extraction	Conditional state and success probability are both specified	Conditional improvement must be distinguished from average resource yield.

5.3 Optimization and finite-resource interpretation

Optimizing temporal profiles can improve the leading-order harvesting gap, but the optimized profile must remain inside the approximation regime. Hermite-basis optimization distinguishes approximately compact, weakly communicating, and causally connected configurations; large gains in a truncated calculation do not establish finite-coupling performance [48]. Circuit-derived variable-gap detectors and Bose-polaron proposals introduce different microscopic constraints [49, 50]. Their relevance is the explicit map from controls to an effective interaction, not a common temperature or numerical witness value.

With detector ensembles, bipartite entanglement between two groups is not automatically genuine multipartite entanglement among all probes [51]. Access to additional field regions and classical records can also change the extractable resource [52]. Such protocols require those operations to be included in the resource budget.

Algebraic embezzlement addresses another limit: suitable local operations can borrow entanglement from a reference state with small state disturbance in the operator-algebraic setting [53, 54]. These results do not specify a finite-energy detector protocol with arbitrary yield.

Here the operational quantity remains the resource produced by a stated probe interaction with controlled physical costs.

6 Resource certification and multipartite diagnostics

6.1 A witness requires a measurement scenario

Entanglement, steering, Bell nonlocality, coherence, magic, and contextuality are defined using different operational structures. In a fixed bipartite scenario,

$$\text{Bell violation} \Longrightarrow \text{steerability} \Longrightarrow \text{entanglement}, \tag{21}$$

while the converses fail in general [55–58]. Coherence depends on a reference basis and free operations; its conversion to entanglement is a theorem within specified resource theories [59–62]. It does not identify detector coherence with negativity.

Magic–contextuality relations likewise depend on dimension, representation, and the allowed subtheory. Contextuality is necessary for computational advantage in the relevant odd-prime stabilizer setting, but arbitrary contextuality is not a sufficient distillation guarantee [63–65]. A contextual fraction refers to a specified empirical measurement model [66].

Table 5: Reference structures needed to interpret detector-resource studies.

Resource	Typical diagnostic	Information needed beyond its numerical value
Entanglement	Negativity, concurrence, entanglement witness	Bipartition, normalization, state reconstruction, and mixed-state criterion.
Steering / Bell nonlocality	Assemblage tests / Bell inequalities	Trusted parties, direction, measurement settings, and statistical confidence.
Coherence	Basis-dependent coherence or robustness	Reference basis, free operations, and conversion or discrimination task.
Magic / contextuality	Mana, stabilizer indicators / contextual fraction	Dimension, stabilizer subtheory, and the full measurement scenario.

Detector models make these requirements concrete. Nonidentical gaps can produce directional steering ranges [67]. Magic-harvesting studies use detector dimensions and stabilizer diagnostics that need not agree with entanglement ordering [68, 69]. The qutrit contextuality protocols of LeMaitre and of Lima *et al.* employ different measurement scenarios, so their contextual fractions are not directly interchangeable [70, 71]. Their noncompact profiles also require causal-tail qualifications. The relevant comparison is the certified property under the actual measurement model, not a ranking across unrelated resource scales.

6.2 Residual negativity does not certify mixed-state GME

A mixed state is biseparable if it is a convex mixture of states separable across possibly different bipartitions. Genuine multipartite entanglement (GME) excludes every such decomposition. Residual combinations of bipartite negativities are not automatically faithful GME indicators on mixed states [72–74].

An explicit diagnostic is

$$\rho_{\mathrm{b}} = \frac{1}{3} \sum_{XY|ZW \in \{AB|CD, AC|BD, AD|BC\}} \Phi^{+}_{XY} \otimes \Phi^{+}_{ZW}, \qquad \Phi^{+} = |\Phi^{+}\rangle\langle\Phi^{+}|. \tag{22}$$

Here $|\Phi^{+}\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. Each component is separable across a $2|2$ cut, so the mixture is biseparable. Nevertheless, with the normalized convention $N = \|\rho^{T_A}\|_1 - 1 = 2\mathcal{N}$, all four $1|3$

negativities equal one and all six pair negativities vanish. Therefore the raw residuals

$$\pi_A = N^2_{A|BCD} - \sum_{X=B,C,D} N^2_{AX} \tag{23}$$

and their analogous aggregate π_4 equal one. A maximal value of this residual does not certify four-party GME. Supplementary Sec. S3 gives the three-party counterpart and the relation to the detector studies; the accompanying script directly diagonalizes the required partial transposes. We use "residual-negativity indicator" unless a mixed-state GME witness or an appropriate extension is separately established.

6.3 Structural invariance and perturbative revival are different claims

For

$$|\mathrm{CL}_4\rangle = \tfrac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle - |1111\rangle), \tag{24}$$

the reduced ABC state already has $N_{A|BC} = N_{B|AC} = 1$. Any local CPTP channel on D leaves that reduced state unchanged. Negativity monotonicity under tracing out D, together with the qubit upper cap, then implies $N_{A|BCD} = N_{B|ACD} = 1$ after the channel. Thus the corresponding freezing in the detector calculation [75] has an exact state-and-channel explanation, independent of a specifically Unruh-induced mechanism.

A different issue arises in finite-order revival plots. Set $q = e^{-2\pi\Omega/a}$ and let ν be the effective coupling. In the first-order W and Dicke models considered in Supplementary Sec. S3, the normalization corrections are

$$\delta_W = \frac{\nu^2(1+3q)}{4(1-q)}, \qquad \delta_D = \frac{\nu^2(1+q)}{2(1-q)}. \tag{25}$$

The reconstructed turning points at the tested couplings occur with corrections of order unity [76, 77]. This demonstrates nonuniformity of those truncations as $q \to 1$, not absence of revival in the exact dynamics. A small normalization correction is a useful necessary diagnostic here, not a sufficient bound on all higher-order errors. The Star-state crossover at moderate q provides a contrasting small-correction example [78]. Source-dependent reconstructions and independently testable state identities are distinguished in the supplement and verification record.

7 Utility test I: a field-mediated channel

A positive detector witness does not specify a communication rate. A channel requires an encoding, a causal transfer interaction, a decoder, and a performance criterion. Covariant qubit–field–qubit constructions express this map through field correlations and the causal propagator [79]; suitably engineered localized-qubit protocols can attain maximal quantum capacity within a specified scalar-field model [80]. This is an existence result for an engineered channel, not a property of arbitrary harvesting couplings.

For a channel $\mathcal{E}$, quantum capacity is given by regularized coherent information,

$$Q(\mathcal{E}) = \lim_{n\to\infty} \frac{1}{n} \max_{\rho_n} I_c(\rho_n, \mathcal{E}^{\otimes n}), \tag{26}$$

with optimization over the allowed inputs and the stated resource constraints [81, 82]. A one-use witness or coherent-information value need not equal this regularized quantity. Classical and entanglement-assisted capacities concern different operational budgets [83, 84].

The causal propagator also determines which geometries can carry an encoded disturbance. Near Schwarzschild backgrounds, direct null paths, orbiting null paths, and timelike tails can contribute differently [85]. Correlated receiver noise may improve joint decoding even when one

receiver alone is outside the sender's causal future [86]; this does not make its individual output a superluminal signal. A multi-receiver protocol therefore needs its full joint output map.

For the present review, the practical test is whether the claimed resource supports the stated input–output task after preparation, interaction, and readout are included. A capacity comparison must match the encoding alphabet, admissible energy, timing, receiver access, and decoder. This criterion keeps communication closely connected to probe dynamics without treating optical-network engineering or gravity-mediated transfer as harvesting evidence.

8 Utility test II: identifiable detector parameters

8.1 Sensitivity and attainable precision

Detector metrology estimates parameters from a specified readout. For outcome probabilities $p(x|\theta)$,

$$F_{\mathrm{cl}}(\theta) = \sum_x \frac{[\partial_\theta p(x|\theta)]^2}{p(x|\theta)}, \qquad \mathrm{Var}(\hat{\theta}) \geq \frac{1}{NF_{\mathrm{cl}}(\theta)}, \tag{27}$$

under the local unbiasedness and regularity conditions of the Cramér–Rao bound. Quantum Fisher information (QFI) optimizes over measurements,

$$\partial_\theta \rho = \tfrac{1}{2}(L_\theta \rho + \rho L_\theta), \qquad F_Q(\theta) = \mathrm{Tr}(\rho L_\theta^2). \tag{28}$$

It is a local sensitivity bound, not a guarantee that an implementable measurement attains it [87–90]. In multiparameter problems, incompatible optimal measurements can prevent joint saturation of the SLD matrix bound; the cost function and measurement class must be specified [90, 91].

Relativistic-metrology frameworks encode parameters in field transformations or detector dynamics [92–94]. Localized detector inference adds dependence on initial state, switching, field state, and the actually measured observable. The same resource witness can therefore coexist with different parameter sensitivities.

8.2 A matched thermal example

For a thermalized two-level detector with known gap Ω and $p_e = (1 + e^{\Omega/T})^{-1}$, population measurement attains the QFI in this diagonal family [95]. The QFIs for absolute and logarithmic temperature are

$$F_T = \frac{\Omega^2}{4T^4}\,\mathrm{sech}^2(\Omega/2T), \qquad F_{\ln T} = T^2 F_T. \tag{29}$$

With $x = \Omega/T$, the relative-precision optimum satisfies $x\tanh(x/2) = 2$, giving $T/\Omega = 0.416778\ldots$; the absolute-temperature optimum satisfies $x\tanh(x/2) = 4$, giving $T/\Omega = 0.242091\ldots$. This changes the precision question, not the detector state. It is an elementary illustration of why parameterization and loss must be fixed before comparing sensitivity [96].

8.3 Identifiability, nuisance parameters, and full-cycle cost

For a parameter of interest θ and nuisance vector $\boldsymbol{\nu}$, the Fisher matrix of an actual readout gives

$$F_{\mathrm{eff}} = F_{\theta\theta} - F_{\theta\nu}F_{\nu\nu}^{-1}F_{\nu\theta}, \qquad \mathrm{Var}(\hat{\theta}) \geq \frac{1}{NF_{\mathrm{eff}}}, \tag{30}$$

when the regular model is locally identifiable and $F_{\nu\nu}$ is invertible. If curvature and acceleration enter only through one effective temperature, one readout cannot identify both without another

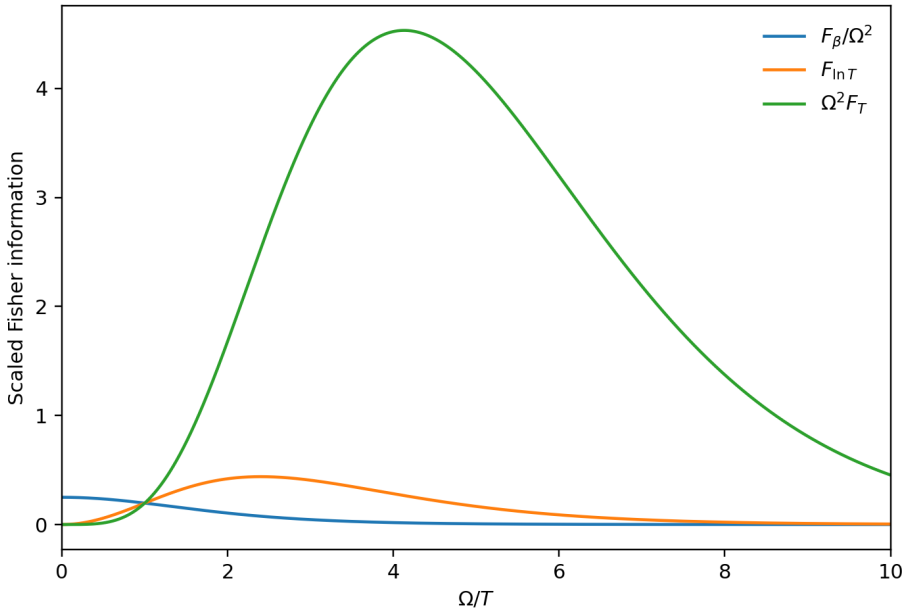


Figure 1: Parameterization-dependent optima in the same thermalized two-level model. The inverse-temperature QFI, logarithmic-temperature QFI, and rescaled absolute-temperature QFI answer different precision questions.

observable or calibration. Coherently controlled trajectories can access additional path correlations in specified models [97], but require additional preparation and interference-sensitive readout.

The repetition count is also constrained by $N \leq t_{\rm tot}/(t_{\rm prep} + t_{\rm int} + t_{\rm read})$ in a sequential experiment. Postselection and ancillary probes change the information per full cycle. De Sitter and acceleration studies illustrate parameter- and state-dependent sensitivities [96, 98]; comparisons should report attainable readout information, nuisance calibrations, success probability, and full-cycle cost, rather than QFI alone.

9 Matching backgrounds and coupled observables

9.1 A common response template is not a universal spectrum

For a Hermitian coupled observable $\mathcal{O}$, the leading response is

$$P_\nu^{(2)} = \lambda_\nu^2 \int d\tau\, d\tau'\, \chi_\nu(\tau)\chi_\nu(\tau') e^{-i\Omega_\nu(\tau-\tau')} W_{\mathcal{O}}(x_\nu(\tau), x_\nu(\tau')), \tag{31}$$

where $W_{\mathcal{O}}(x,x') = \langle \mathcal{O}(x)\mathcal{O}(x')\rangle$. Extended probes add spatial integrals. For two probes the same model must supply both local responses and the cross-correlator. Matching gaps or KMS temperatures does not match these kernels [99–105].

9.2 Scalar mass, curvature coupling, and the sampled region

The equation $(\Box - m^2 - \xi R)\phi = 0$ depends on both mass and curvature coupling. In Schwarzschild spacetime $R = 0$, so changing ξ alone leaves this equation unchanged. A massive scalar instead changes the radial potential,

$$V_\ell(r) = \left(1 - \frac{2M}{r}\right)\left[m^2 + \frac{\ell(\ell+1)}{r^2} + \frac{2M}{r^3}\right]. \tag{32}$$

Table 6: Representative probe couplings and the controls needed for comparison.

Field	Coupled observable	Required controls
Real scalar	ϕ or a proper-time derivative	Mass, curvature coupling, state, infrared prescription, switching, and spatial profile.
Dirac	Even composite : $\bar{\psi}\psi$: for the bosonic probe considered here	Spin connection, parallel transport, composite-operator prescription, and field state.
Maxwell	Electric dipole $d^i E_i$ or field strength	Gauge-invariant tensor kernel, tetrads, polarization, orientation, atomic form factors, and boundary conditions.

The 'in' sector starts at $\omega = m$, while horizon-originating 'up' modes exist below that asymptotic threshold and decay at infinity. A static detector at r_0 samples $\omega = \Omega\sqrt{1-2M/r_0}$; it may therefore respond to thermally occupied up modes below m. The long-range Coulomb tail also matters for radial normalization [106, 107]. Supplementary Sec. S10 retains a local-response benchmark, without interpreting its spectral enhancement as two-detector harvesting.

The cross-correlator must preserve the phases of the radial products. For one sector write $C_{AB} = u(r_A)u^*(r_B)$, including its real normalization factors. The corresponding integrand is

$$(1+n_\omega)C_{AB}e^{-i\omega\Delta t} + n_\omega C^*_{AB}e^{+i\omega\Delta t}. \tag{33}$$

The conjugation follows from $\langle aa^\dagger\rangle = 1+n_\omega$ and $\langle a^\dagger a\rangle = n_\omega$. It ensures that the commutator is independent of the occupation number. A common complex factor for both terms is generally incorrect; local $r_A = r_B$ spectra are a real-valued special case. This consistency condition matters before extending a local response calculation to harvesting.

In de Sitter spacetime, $R = 12H^2$ and the Bunch–Davies index is

$$\nu = \sqrt{\frac{9}{4} - \frac{m^2}{H^2} - 12\xi}. \tag{34}$$

Equal detailed-balance temperatures can coexist with different infrared spectra and switching terms [108, 109]. Derivative coupling samples $\partial_\tau\partial_{\tau'}W$; integration by parts changes the effective switching and requires the relevant boundary terms to vanish. It can remove an additive infrared constant in two dimensions without making the derivative and monopole models equivalent [110, 111]. Supplementary Sec. S8 gives the response identities.

9.3 Horizons and nonlocal correlations

Black-hole calculations must distinguish local excitation, selected interior–exterior mode entanglement, and the state of two localized probes. Freely falling derivative-coupled detectors in a dimensionally reduced model and static probes in four dimensions sample different states and kernels [112–114]. In four-dimensional Schwarzschild calculations, lensing and caustics can change whether the nonlocal amplitude is dominated by commutator or anticommutator contributions [115]. Noncompact switching remains a separate causal qualification. No universal horizon-induced enhancement or suppression follows from these distinct models.

The same caution applies to observer-dependent subsystem choices. A global Lorentz transformation can change a reduced spin state after momentum is discarded [1]. Such redistribution of accessible correlations is distinct from a localized field-to-detector extraction protocol. Comparisons in this review keep the measured subsystem explicit.

9.4 Dirac and electromagnetic probes

For an ordinary bosonic detector, a linear coupling to a fermionic field does not preserve fermion parity. The scalar-density model instead uses the even observable : $\bar{\psi}\psi$:, whose two-point

function is a composite four-field correlator. Spin connection, parallel transport, and renormalization prescriptions cannot be supplied by substituting a scalar Wightman function by analogy [116, 117]. Detectors with fermionic degrees of freedom are a different construction. The model-specific identities are retained in Supplementary Sec. S9.

An electromagnetic dipole probes a tensor correlator. Before atomic form factors are inserted, the transverse-projector angular integral separates into perpendicular and parallel contractions relative to the separation vector:

$$A_{\perp}(z) = \frac{4\pi}{3}[2j_0(z) - j_2(z)], \qquad A_{\parallel}(z) = \frac{8\pi}{3}[j_0(z) + j_2(z)], \quad z = kR. \tag{35}$$

Both equal $8\pi/3$ at $z = 0$, but differ at finite separation. The correct prefactors and assignment to orientations are essential. Supplementary Sec. S4 gives the decomposition, and the accompanying check compares it with direct angular quadrature. Atomic anisotropy and angular-momentum selection add further structure [118]; the vector-field label alone predicts no universal harvesting advantage.

Gauge-invariant probe response does not require assigning a tensor factor to a field region. Regional entanglement does require a local-algebra and boundary prescription. Gauss constraints, centres, and edge degrees of freedom must therefore be fixed before using field-region resource bounds [119–121]. These requirements complement the detector-side matching of preparation, gap, switching, smearing, normalization, and witness.

10 Experimental evidence and a certification agenda

10.1 What the measured observable establishes

Electro-optic sampling (EOS) connects optical probes to terahertz field correlations. A detector-state entanglement witness has been proposed within this platform [122]. The vacuum/source-correlation experiment reported by Herter *et al.* resolves contributions associated with vacuum fluctuations and source radiation [123]. The measured correlations are an important field-sampling result; they do not by themselves certify entanglement of the reduced probe state or a Bell violation. The next inference requires the observables and calibrations of an appropriate probe-state witness.

Microscopically motivated detector proposals supply complementary evidence. Superconducting variable-gap models relate circuit controls to a derivative-coupled detector interaction [49]. The Bose-polaron proposal maps trapped impurities to probes of an effective phonon field and estimates achievable witness strength and measurement demands [50]. These are implementation proposals. An analogue field and a relativistic vacuum are different physical inputs, even when part of the detector dynamics is reproduced.

10.2 A minimal reproducible harvesting comparison

A useful report fixes the field state and coupled observable; detector dimensions, preparation, gaps, switching and smearing; and complete interaction supports. Local probabilities and nonlocal amplitudes must use the same regularization and perturbative order. Numerical error, omitted-order error, and apparatus uncertainty are different contributions and should be recorded separately. Agreement of two integration settings is a convergence diagnostic, not a certified total error bound.

The worked examples in the supplement serve different checks. The finite-mode JC sums and the independent compact-support Fourier reconstruction verify selected algebraic and integration steps. The multipartite counterexamples test what a witness certifies. The Schwarzschild asymptotic diagnostic checks boundary expansions, not a full radial scattering or harvesting calculation. Their verification record does not certify every source-dependent reconstruction,

Table 7: Evidence and the remaining step toward harvesting certification.

Platform / result	Evidence supplied	Additional requirement
EOS vacuum/source sampling [123]	Time-resolved optical field-correlation measurements	Probe-state entanglement witness with calibrated readout and uncertainty.
EOS harvesting design [122]	A platform-specific proposed witness	Experimental implementation and a positive statistically certified witness.
Variable-gap superconducting probes [49]	Circuit-derived interaction model and theoretical dynamics	Device calibration, controlled approximations, and a causal-support or signalling-error budget.
Bose-polaron probes [50]	Effective-field mapping and feasibility estimates	Validity of the mapping, loss and sampling analysis, and measured nonlocal coherence.

de Sitter integral, or radial solution. A small positive $O(\lambda^2)$ harvesting coefficient still requires higher-order control before being interpreted as an experimentally resolvable yield.

10.3 Synthesis and priorities

The comparisons support three conclusions. First, a common probability is not a common instrument: matching the readout and back-action is needed to connect measurement formalisms. Second, accessible field correlation is an input budget, not a guaranteed extracted resource: the channel and complete spacetime support determine what reaches the probes. Third, a resource indicator is not an operational advantage: mixed-state certification and an attainable task benchmark are separate requirements.

These conclusions lead to four concrete priorities:

1. Compare two measurement formalisms on one compact-support probe experiment, reporting both probabilities and conditional updates.
2. Publish harvesting boundaries together with causal boundaries and explicit approximation diagnostics, including higher-order checks near small witness gaps.
3. Match scalar, Dirac, and electromagnetic models at the level of the coupled observable and physical controls, and state residual mismatches.
4. Design experimental readout around a robust probe-state witness, success probability, and full-cycle communication or estimation performance.

These are testable comparisons that can connect existing model calculations, rather than a requirement to collapse all resources or backgrounds into a single numerical hierarchy.

11 Conclusions

Localized probes connect quantum-field measurements to finite resource-bearing systems, but the connection has several independently testable stages. This review has organized selected literature around causal implementability, interaction support, resource certification, and attainable tasks. Its worked examples distinguish geometric localization from shrinking spatial size, raw multipartite residuals from GME, and structural local-channel invariance from a background-specific mechanism.

The resulting comparisons sharpen both positive results and their limits. Compact spacelike interactions can support a positive leading harvesting gap in a stated model; regional mutual information bounds do not supply its yield. Tensor and composite-operator kernels permit comparisons beyond scalar monopole coupling, but shared temperatures do not make those kernels equal. Optical correlation measurements supply experimental access to relevant field observables, while probe-state certification remains a further step. Progress now depends on

matched physical models, controlled errors, and readouts that certify the resource and task actually claimed.

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